

SQL Constraints: CHECK and DEFAULT

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CHECK Constraint

A CHECK constraint ensures that inserted or updated data satisfies a specified Boolean condition.

```
CREATE TABLE Employees(  
    EmployeeID INT PRIMARY KEY,  
    Salary DECIMAL(10,2),  
    CHECK (Salary > 0)  
);
```

Now:

```
Salary = -50000
```

```
CHECK (Percentage BETWEEN 0 AND 100)
```

```
CHECK (Quantity >= 0)
```

Complex Conditions

A CHECK expression can combine multiple conditions.

```
CHECK (  
    Department='ECE'  
    OR  
    Department='CSE'  
)
```

Multi-Column CHECK

One of the biggest advantages of table-level CHECK.

```
CREATE TABLE Projects(  
    StartDate DATE,  
    EndDate DATE,
```

```
CHECK (StartDate < EndDate)
);
```

CHECK with Functions

Many databases allow functions inside CHECK expressions.

```
CHECK (
    CHAR_LENGTH(Name) >= 3
)
```

```
CHECK (
    YEAR(BirthDate) >= 1900
)
```

Vendor Limitations

This is an interview favorite.

- **MySQL**
- Versions before 8.0.16 parsed CHECK constraints but ignored them.
- From 8.0.16 onward, CHECK constraints are enforced.

Internal Flow



Note: CHECK constraints are evaluated only during INSERT and UPDATE. They do not slow down normal SELECT queries.

DEFAULT Constraint

A DEFAULT constraint supplies a predefined value when no value is explicitly provided for a column.

Syntax

```
CREATE TABLE Users(  
    UserID INT PRIMARY KEY,  
    Country VARCHAR(50)
```

```
        DEFAULT 'India'  
    );
```

CURRENT_DATE

```
JoiningDate DATE DEFAULT (CURRENT_DATE)
```

CURRENT_TIMESTAMP

```
CreatedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP
```

This is one of the most common defaults used in production.

MySQL does not allow arbitrary functions like `UUID()` directly as a column DEFAULT. A common approach is:

```
INSERT INTO Users(UserID) VALUES (UUID());
```

ALTER DEFAULT

```
ALTER TABLE Users  
MODIFY Country VARCHAR(50) DEFAULT 'USA';
```

Remove Default

```
ALTER TABLE Users  
MODIFY Country VARCHAR(50);
```

If NULL is explicitly inserted, the default is not used.